and later Trivers and Hare, on to one of the most spectacular triumphs of the selfish gene theory. The reasoning goes like this.

Insects of the group known as the Hymenoptera, including ants, bees, and wasps, have a very odd system of sex determination. Termites do not belong to this group and they do not share the same peculiarity. A hymenopteran nest typically has only one mature queen. She made one mating flight when young and stored up the sperms for the rest of her long life—ten years or even longer. She rations the sperms out to her eggs over the years, allowing the eggs to be fertilized as they pass out through her tubes. But not all the eggs are fertilized. The unfertilized ones develop into males. A male therefore has no father, and all the cells of his body contain just a single set of chromosomes (all obtained from his mother) instead of a double set (one from the father and one from the mother) as in ourselves. In terms of the analogy of Chapter 3, a male hymenopteran has only one copy of each 'volume' in each of his cells, instead of the usual two.

A female hymenopteran, on the other hand, is normal in that she does have a father, and she has the usual double set of chromosomes in each of her body cells. Whether a female develops into a worker or a queen depends not on her genes but on how she is brought up. That is to say, each female has a complete set of queen-making genes, and a complete set of worker-making genes (or, rather, sets of genes for making each specialized caste of worker, soldier, etc.). Which set of genes is 'turned on' depends on how the female is reared, in particular on the food she receives.

Although there are many complications, this is essentially how things are. We do not know why this extraordinary system of sexual reproduction evolved. No doubt there were good reasons, but for the moment we must just treat it as a curious fact about the Hymenoptera. Whatever the original reason for the oddity, it plays havoc with Chapter 6's neat rules for calculating relatedness. It means that the sperms of a single male, instead of all being different as they are in ourselves, are all exactly the same. A male has only a single set of genes in each of his body cells, not a double set. Every sperm must therefore receive the full set of genes rather than a 50 per cent sample, and all sperms from a given male are therefore identical. Let us now try to calculate the relatedness between a mother and son. If a male is known to possess a gene *A*, what are the chances that his mother shares it? The answer must be 100 per